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Matilda HPC Cluster



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Running fMRIPrep

Hardware Available

- A Linux-based cluster for workloads that do not fit on an individual computer.
- More than 2,000 CPU cores across standard, high-throughput, big-memory, hybrid, and GPU nodes.
- All nodes are connected by 100 Gbps HDR100 InfiniBand.
- Buy-in nodes are funded by a specific department or research group. We can use them only if our PI or group has access.

- **Standard compute:** 40 nodes; 40 CPU cores and 192 GB RAM per node.
- **High throughput:** 10 nodes; 8 CPU cores and 192 GB RAM per node.
- **Big memory:** 4 nodes; 40 CPU cores and 768 GB RAM per node.
- **GPU:** 3 nodes; 4 NVIDIA Tesla V100 16 GB GPUs, 48 CPU cores, and 192 GB RAM per node.
- **Specialized buy-in nodes:**
  - Chemistry: 1 node with 1.5 TB RAM.
  - SECS: 2 nodes with NVIDIA RTX Pro 6000 Blackwell GPUs and 512 GB RAM.
  - Availability depends on PI or research-group allocation; they are not automatically available to every user.

Storage and File Structure

- **Home:** personal files and configuration.
- **Projects:** shared research data and collaboration.
- **Scratch:** fast temporary working space for jobs.
- Home and project storage is backed up; scratch is not.

`/home/m/moghaddami:` 50 GB per user; backed up and replicated

`/projects/[PI NetID]/[Project Name]` 1 TB per project; shared with project members

`/scratch/users/moghaddami` 10 TB per user; 100 Gbps InfiniBand

`/scratch/projects/adresearch` 10 TB per project; shared with group members

- Scratch files are automatically deleted after 45 days.

Thank you!